Avi Patel

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Education

University of Victoria — Victoria, BC
Bachelor of Science in Physics, Minor in Computer Science — *Sept. 2019 – June 2025*

Stanford University — Coursera
Machine Learning Certificate — *Jan. 2022*

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R, LaTeX
Tools: Git, VS Code, MS Office
Libraries: pandas, NumPy, Matplotlib
Communication: Technical Research/communication

Experience

Dough Master/Team member — 2026 – curr
Pizza Hut — *Welland, ON*

- Recipe Accuracy: Weigh and measure ingredients precisely, and monitor water temperatures to ensure consistent dough quality.
- Dough Preparation: Mix, portion, and pan dough sizes according to specifications.
- Proofing: Monitor dough as it rises to achieve the ideal height and size
- Sanitation: Keep preparation areas, work surfaces, and utensils clean and safe.

E-Commerce operations assistant — 2025 – 2026
SFUMATO (maisonsfumato.com) — *remote*

- Maintained product catalog, pricing, and inventory data on Shopify
- Updated product listings and kept digital records accurate and organized
- Supported daily operations of a small DTC retail business

Delivery Expert & Customer Service — 2022 – 2025
Dominos — *Victoria, BC*

- Processed high volumes of financial transactions with 100% accuracy, including cash and digital payments, ensuring accuracy and compliance with company procedures
- Delivered consistent, high-quality customer service to 50+ customers per shift, ensuring positive experiences
- Identified customer needs and provided tailored solutions, contributing to increased customer satisfaction and repeat business
- Maintained 98% on-time service performance through effective time management and prioritization

Academic Projects / Work

Rocketry Club team member | *Experimental Design, Collaboration, Python, Testing, SolidWorks* — 2019-2021

- Worked alongside 10-20 team members to design and build model rocket for competition
- Modeled sub-team rocket components in SolidWorks
- Created and Edited documentation for parts of rocket and design
- Collaborated on creating rocket simulation using python

Van der Graaf Generator | *Electronics, Experimental Design, Prototyping, Electrostatics* — April 2023

- Collaborated in a 4-person team to design and construct a functional Van de Graaff generator for an upper-year physics project
- Interfaced physical components (belt, electrodes, motor) and tuned discharge behavior through iterative testing
- Presented findings to peers and faculty, receiving top marks for experimental setup and technical communication

Scientific Lab Work | *Data Analysis, Python, Excel, Technical Writing* — 2019 – 2025

- Authored 80+ lab reports across diverse physics domains including optics, thermodynamics, electromagnetism, and astrophysics

- Conducted rigorous experimental data acquisition and applied statistical methods to assess uncertainty, validate hypotheses, and interpret physical systems
- Developed clear, technically sound reports tailored to academic audiences, demonstrating precision in communicating complex methodologies and outcomes

Patient Management System | *Python, MVC Architecture, GUI, Git, Unit Testing* Sept. – Dec. 2024

- Designed and implemented a modular healthcare application in Python using the Model-View-Controller (MVC) architecture to separate business logic, data management, and user interface
- Developed full CRUD functionality for managing patient records, including validation, persistent storage, and dynamic updates
- Built an interactive graphical user interface (GUI) to simulate real-world clinical workflows and improve user accessibility
- Collaborated in a two-person team across multiple development cycles; applied version control and software engineering best practices throughout

Experimental Physics | *Circuit Design, Signal Processing, Instrumentation* Jan. 2022

- Designed and assembled small-scale analog circuits including RC filters, audio amplifiers, and a sound-to-light converter using breadboards and discrete components
- Utilized oscilloscopes and multimeters to test signal behavior, measure frequency response, and validate theoretical predictions
- Interpreted schematics, calculated component values, and debugged circuit behavior through iterative testing and analysis
- Reinforced core electronics principles by translating theoretical knowledge into practical prototypes and experimental validation

Ray Tracer | *Python, File Parsing, Image Rendering, Computer Graphics* Jan. 2025 – Apr. 2025

- Implemented a ray tracing algorithm in Python as part of a computer graphics course to render 3D scenes from textual scene descriptions
- Developed a file parser to interpret object positions, materials, light sources, and camera settings from a custom input format
- Calculated light paths, shading, and reflections using vector math and intersection algorithms; generated final scenes as .png images
- Demonstrated proficiency in geometric transformations, recursive algorithms, and pixel-level image synthesis

Physics Seminar Presentations | *Technical Research, Science Communication* Jan. 2023 – Apr. 2023

- Researched advanced physics topics including optical black hole analogues and the physics behind nuclear weapons, synthesizing academic literature and theoretical models
- Created and delivered technical presentations tailored for peer and faculty audiences, translating complex physical principles into accessible formats
- Demonstrated strong scientific communication, slide design, and public speaking skills in a formal academic setting

Soft Skills

Analytical Thinking: Applied to complex physics & mathematics problems
Collaboration: Worked in teams on projects and labs
Scientific Communication: Delivered 80+ technical reports and physics presentations
Time Management: Balanced coursework, projects, and work over 3 years
Adaptability : Self-taught machine learning and statistics, applying it to real data
Customer Service : Maintained professionalism over 1000+ Domino's shifts